EARLY-CHILDHOOD AUTISM SCREENING VIA RANDOM FOREST

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ABSTRACT

We are interested in applying random forest models to classify risk of autism spectrum disorder in toddlers. We will leverage a large community-convenience sample ($N = 3727$) collected from over 18 clinical instruments ($n_{\text{features}} = 5204$) across four longitudinal timepoints. This problem has not been approached with machine learning for such a comprehensive clinical battery. Our aim is to appropriately build random forest models that are modified and sensitive to (1) heterogeneous behavioral-phenotypes characteristic of ASD and (2) class imbalances in diagnostic outcomes. We address these issues by optimizing oversampling techniques like SMOTE, BorderlineSMOTE, and ADASYN; implementing feature selection via principle component analysis; and improve the random forest ensemble method by implementing a gradient-boosted random forest model. Results suggest for this context, a combination of ADASYN and gradient boosted models are most appropriately suited to address the class imbalances and heterogeneity of the dataset, and have an overall F1 score of 0.95 with ROC AUC 0.97.

1 INTRODUCTION

Autism spectrum disorder (ASD) has been researched for decades yet is still notably challenging to diagnose due to considerable heterogeneity within behavioral, genetic, neurological, and physiological profiles. Clinical research suggests that early diagnosis (in the first 3 years of life) is critical to improving outcomes for moderate to severe symptomatology. We are interested in applying classical machine learning techniques (support vector machines) to classify early likelihood (or risk) and clinical diagnosis of autism using a comprehensive clinical battery.

While the aim of leveraging machine learning to autism diagnosis is not novel, there has been limited success due to three primary factors. First, ASD has wide heterogeneity in behavioral outcomes (i.e., ASD presents differently in individuals despite having similar impairments) leading to low confidence in machine learning classification and clustering. Second, ASD has relatively low prevalence (1 in 32 for 8 year-olds, and 1 in 53 for 4 year-olds), causing many datasets to have class imbalances that impact classification outcomes for models. Third, many large datasets have sparse or incomplete clinical batteries leading to many missing values, as these datasets are compiled across disparate studies with a variety of goals.

As a unique use of machine learning on a particularly extensive dataset, this project aims to categorize toddlers' risk for autism spectrum disorder (ASD). This project makes use of support vector machines, as well as fine-tuning of the model to tackle a number of difficult technical and clinical problems. Because of these difficulties, this project is interesting for both the domains of developmental health and machine learning. The project's significance and interest are primarily due to the following factors:

- **Parsing heterogeneity with extensive clinical dataset:** With 5204 features gathered from more than 18 clinical instruments and longitudinal data from 3727 participants, the data set used in this project is one of the most extensive. ASD datasets rarely have complete comprehensive data of this depth and scope. This presents an opportunity for applications of machine learning techniques to find subtle patterns and risk indicators that are difficult to find using conventional techniques. Further, it goes beyond existing machine learning models that are limited to 2-5 clinical instruments available for their samples.

- **Resolving class imbalances:** The dataset shows that the proportion of children classified as LL ($N_{\text{LL}} = 1799$) is significantly higher than the HL group ($N_{\text{HL}} = 1228$); this is representative of the relatively low prevalence of ASD in the general population. Because machine learning algorithms have a tendency to favor the majority class, class imbalance presents a significant challenge. This results in biased models that are less sensitive to predictions from minority classes. The goal of this project is to improve the dependability of ASD risk classification by improving SVM models to account for known class imbalances.
- **Novelty in ASD prediction:** Although machine learning has been used to predict ASD in older populations, this project addresses this issue with clinical data for young children. Although early detection and intervention is critical, this age group has been underrepresented in machine learning-based models of ASD classification due to the difficulty of early developmental assessments and the scarcity of data. This project seeks to improve domain-specific machine learning methods in pediatric health by concentrating on this group and leveraging a unique and rich dataset.

1.1 Prior Work

Previous studies have shown that machine learning methods can be used to identify autism spectrum disorder across the lifespan. Mahedy Hasan et al. (2023) created a comprehensive ML framework that includes classifiers like support vector machines, decision trees, and random forest models. Using feature scaling and selection techniques, the framework achieved high accuracy of 98%. Yuvaraj et al. (2024) addressed issues such as the scarcity of data sets and the imbalance between classes using SVM, logistic regression, and naive Bayes to investigate behavioral and personal traits in their study to detect ASD in the adults age group. There is one study that focused on the toddler age group. Islam et al. (2020) utilized RF and K-Nearest Neighbors classifiers for toddlers, as well as the potential of mobile-based tools for early ASD detection. In these studies, class imbalance has been mentioned as a challenge, and ReliefF and correlation-based selection have been crucial to addressing it.

There are still restrictions in place despite these developments, especially with regard to the use of small datasets. In contrast, this project will address said limitations by utilizing a large, longitudinal dataset and fine-tuning RF models for high-dimensional data, while aiming to maintain a high accuracy achieved by the models in previous studies. In more recent studies, the Random Forest classifier was applied to estimate autism risk in adults and children. The paper published by Gill et al. (2024) focused on implementing a RF-based classifier for estimating ASD risk in children using an extensive dataset incorporating clinical, genetic, and demographic information. In their methodologies, class imbalance was addressed by oversampling to mitigate bias towards the majority class.

Cross validation and confusion matrix analysis was used to judge the model's robustness. Additionally, recent works by Sonoda (2023) employed various oversampling methods to address class imbalance of heterogeneous samples. These methods include SMOTE and its variants such as Borderline-SMOTE and ADASYN, all are effective in various scenarios when applied to samples with class imbalances.

1.2 Data set

Diagnosing autism spectrum disorder (ASD) is challenging due to its low prevalence (1:32) and heterogeneous symptomatology. Autism is defined by three core traits: social communication (SC), restricted behaviors (RRB), and daily living skills (DLS) (IBIS network et al., 2015). Early detection is crucial, as intervention can provide significant long-term benefits (Okoye et al., 2023). However, data for infants and toddlers (ages 0-3) are limited and often lack clinical depth.

The NIH-funded Phenoscreening project collected a community-convenience sample ($N = 3727$) of children aged 18–30 months (Doyle et al., 2021). Families completed seven clinical surveys at 18 and 24 months, with higher-risk children undergoing 15 additional ASD-specific assessments. High-likelihood (HL) and low-likelihood (LL) ASD classifications were derived from survey scores, with HL cases receiving diagnostic evaluations to confirm ASD.

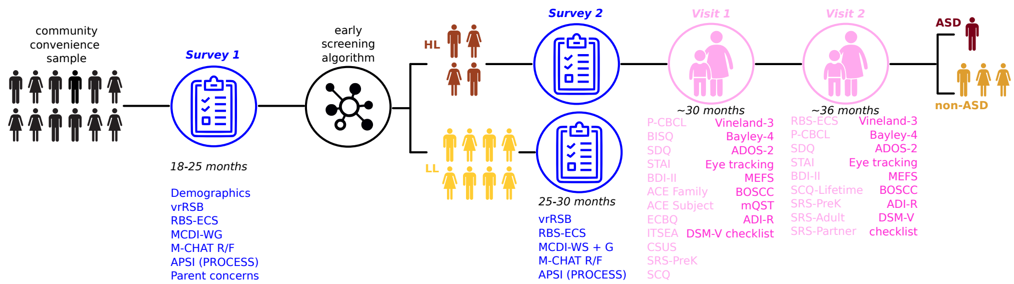


Figure 1: Population Phenoscreening phases of data collection and instruments

Data was collected across four different timepoints, as shown in Figure 1. Parent survey measures are directly shown in blue, and in light pink. Behavioral observation or clinician-collected measures are shown in dark pink. Overall, we aim to reproduce the classifications (HL/LL) from the early screening algorithm for data collected at any timepoint.

1.3 Project Aims

Our project aims to apply and optimize the random forest technique to classify and detect early autism spectrum disorder (ASD) in toddlers—a novel application. Our focus is on addressing two key challenges: (1) heterogeneity, reflecting the variability of ASD symptoms for individuals in terms of behavioral, genetic, neurological, and physiological profiles and (2) class imbalance, where the dataset contains significantly more low-likelihood (LL) cases than high-likelihood (HL) cases, and By tackling these challenges, we aim to improve the performance and reliability of RF for classification and screening of ASD in early childhood.

2 Methodology

2.1 Data cleaning

We completed cleaning and preprocessing the data. We start to inspect the class label to identify imbalances between the LL and HL categories. There are 1799 LL, 1228 HL, and 1692 unknown. For our purposes, the 1692 unknown values are from a pilot study, which only have a single timepoint, and are almost entirely separable based on t-SNE. For this reason, we chose to filter all unknowns from the dataset, and only use the HL/LL labeled data. As for the feature data, the text-based features (enumerated values from surveys) were labeled and encoded, and missing numeric data was replaced with mean values from the distribution of that feature. The cleaned data frame contains 4719 samples and 220 features.

2.2 Random forest models

Random forest is an ensemble method that aggregates predictions from multiple decision tree models, which typically leads to a more generalizable prediction for unseen data. Because they are based on decision trees, which are easily interpretable for how features contribute to a prediction, random forests are an appropriate model for clinical data which require a high degree of interpretability and scrutiny when applied to clinical outcomes (like an ASD diagnosis). We have chosen to implement the out-of-the-box `RandomForestClassifier` as a comparison to a custom random forest implementation, which highlights a few key features of the random forest algorithm.

The custom random forest implementation leverages the sklearn `DecisionTreeClassifier` as its base model. We parallelize the training of multiple trees with a `ThreadPoolExecutor`, since each decision tree can be fit independently. After each tree has been fit, we get predictions from each model and do a majority vote for final predictions. From the documentation, this is different from the sklearn implementation of `RandomForestClassifier`, which uses a soft majority approach (which preserves confidence of prediction from each model). Likely, there is benefit from maintaining the confidence of each prediction; however, because the outcomes for this dataset are very heterogeneous we would expect confidences for many HL points to be closer to chance (i.e., we expect there not to be a strong predictor in many cases).

In our custom implementation, we enabled direct control of `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features` in our `DecisionTreeClassifier`. We directly optimize these parameters, in addition to `n_estimators` which is the variable for the number of decision trees that are modeled, via grid search to tune the hyperparameters of the model.

2.3 Sampling methods for imbalanced classes

Overall, we have a class imbalance between 1799 LL and 1228 HL observations, so we have roughly $1.5\times$ more LL cases. We considered a few different options to oversample our data. First was to use SMOTE, the synthetic minority over-sampling technique, which creates synthetic samples for the minority class based on k-nearest neighbors. Thus it optimizes the full span of the minority class when creating samples.

There is also a variation called BorderlineSMOTE, which avoids oversampling the minority class in regions that are hard to classify, i.e. points on the HL/LL borders in our dataset. We hypothesize that this method may be best for this dataset due to the heterogeneity of the sample, and if we focus on regions with the easiest to separate, i.e. have the least heterogeneity, we will improve overall classification accuracy.

A third method we considered was ADASYN, or adaptive synthetic sampling, which similarly takes k-nearest neighbors, but instead uses impurity ratios of its neighbors to weight the oversampling. We expect this method to optimize oversampling in regions with the most heterogeneity (i.e., in regions that are most difficult to classify). This method balances the utility of SMOTE and BorderlineSMOTE, but uses a softer margin when considering points that are difficult to classify. We hypothesize that this approach will lead to the highest classification performance because it is specifically oversampling for points in regions with high variability between classes.

2.4 Dimensionality reduction

We consider our overall reduced dataset of 220 key features to be the baseline for clinical judgement. However, because instruments could be collected on different dates, we included candidate age for each instrument in this set of 220 features. Furthermore, there is reasonable overlap in the underlying behaviors that the subscores from different instruments measure. For this reason this set of 220 features and subscores may not be purely independent. Thus we use a principal components analysis to reduce dimensionality.

2.5 Improving modeling algorithms

Finally, to improve upon the actual random forest model for the problem-specific case, we apply gradient boosting to random forest methods. As described above, we were able to parallelize our random forest implementation with a `ThreadPoolExecutor` because all decision tree models could be created independently and had no relationship to one another (except for the training data used). Gradient boosting, on the other hand, instead builds trees sequentially, where each new tree helps correct errors made from the previous classifications. This method has been particularly useful for anomaly detection in supervised settings. Because our dataset is highly heterogeneous with large variation of behavioral phenotypes in the HL class, we expect there to be anomalies in the different clusters of phenotypes. We hypothesize optimizing models to improve on misclassifications will be beneficial to addressing heterogeneity.

2.6 Comparison metrics

Model performance was evaluated using standard classification metrics computed from confusion matrices, which tabulate true positives, true negatives, false positives, and false negatives for each class. The F1 score, calculated as the harmonic mean of precision and recall, provided a balanced assessment of model accuracy for the majority class, while the area under the Receiver Operating

Characteristic curve (ROC AUC) offered complementary insights by evaluating performance across all possible classification thresholds. While F1 scores focus on performance at a single decision threshold, ROC AUC provides a more comprehensive view of the model's discriminative ability across the full range of possible threshold values, making these metrics particularly valuable when used in conjunction.

3 RESULTS

Comparison of results from the sklearn `RandomForestClassifer` and our random forest implementation `CustomRandomForest`, shows slight improvements to the model. Using `n_estimators` $= 100$, with `max_depth` $= 10$, `CustomRandomForest` slightly improves classification via (F1 $= 0.936$) compared to the `RandomForestClassifier` (F1 $= 0.935$). However, these models differ by only one sample for the true positive class and are nearly identical.

Implementation of the three oversampling methods show similar small improvements to classification, with ADASYN having the highest performance (F1 $= 0.939$), followed by SMOTE (F1 $= 0.938$) and BorderlineSMOTE (F1 $= 0.926$). Full comparison of results are presented in Table 1.

Table 1: Impact of Oversampling Methods and Dimensionality Reduction on Classification Performance of `CustomRandomForest`

	TP	FP	TN	FN	F1	ROC AUC
sklearn `RandomForestClassifier`	256	51	440	10	0.935	0.974
`CustomRandomForest`	257	50	440	10	0.936	**0.975**
with SMOTE	261	46	438	12	0.938	0.974
with BorderlineSMOTE	247	46	440	10	0.926	0.972
with ADASYN	**262**	**45**	**438**	**12**	**0.939**	0.973
with PCA	197	110	443	7	0.883	0.924
with PCA and SMOTE	229	78	421	29	0.887	0.936

Using PCA while retaining 95% of the variance, 220 features were reduced to 50 components. The reduced dimension data was used as input to the `CustomRandomForest`, with and without oversampling. From Table 1 PCA alone (F1 $= 0.883$) performed slightly worse than PCA with SMOTE oversampling (F1 $= 0.887$). For Random Forest without resampling, PCA reduced the false negatives rate, but increased false positives rate. PCA reduced the computational complexity, but in turn resulted in overall less accuracy and precision.

We implemented the sklearn `RandomForestClassifier` to train and evaluate the dataset, and compare to our custom implementation `CustomRandomForest`.

Our preliminary attempts at implementing gradient boosting are reflected in our `CustomGradientBoosting` model. This directly implements derivations discussed in class, and uses a `SimpleLinearRegressor` as its base model. We use a simple residual $\ell = y - f$ as our loss function. This model underperformed (F1 $= 0.887$) compared to both `CustomRandomForest` (F1 = 0.939) and `RandomForestClassifier` (F1 = 0.936).

We iterated on this implementation of gradient boosting, and specifically improved three aspects. First, we replaced the `SimpleLinearRegressor` base classifier with `DecisionTreeClassifier`. This makes it expliclity a gradient boosted random forest model; this change is motivated by the direct comparison to `CustomRandomForest` as well as the fact that decision trees are often more robust in gradient boosting since they better capture non-linear relations between many different features. Second, we updated the loss function to use log loss, $\ell(y, p) = -[y \log(p) + (1 - y) \log(1 - p)]$, where y is the true label and p is the predicted probability. Note, our implementation uses raw predictions in the log-odds space, so we transform our probabilities with a sigmoid function, $p = \sigma(f) = \frac{1}{1+e^{-f}}$. Third, we add stochastic gradient boosting by subsampling the data, which tends to improve overfitting and improve generalization of the models.

All model hyperparameters were optimized using grid search with stratified 5-fold cross-validation, for `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, and `learning_rate`. Note, gradient boosting models only used `n_estimators`, `max_depth`, and `learning_rate`. The best hyperparameters found for each model are reported in Table 2. Interestingly, `n_estimators` varies considerably between models, but the depth of min-split may also be the main factors that differentiate the models.

Table 2: Best Hyperparameters for Custom Random Forest Model and Gradient Boosting

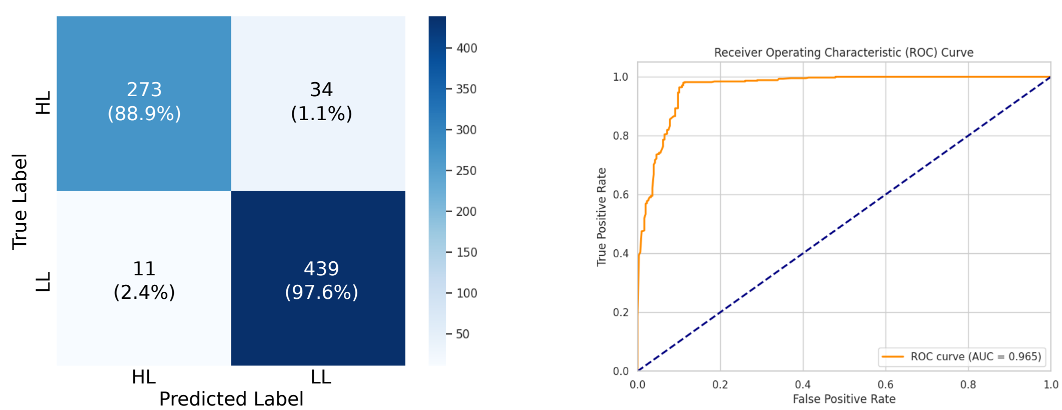
	n_estimators	max_depth	min_samples_split	min_samples_leaf	learning_rate
sklearn `RandomForestClassifier`	90	10	2	4	—
`CustomRandomForest` with ADASYN	30	10	2	1	—
sklearn `GradientBoostingClassifier`	60	5	10	4	
`CustomGradientBoosting` with ADASYN	30	1	—	—	0.01
`ImprovedGradientBoosting` with ADASYN	100	10	—	—	0.1

The `CustomGradientBoosting` model underperformed ($F1 = 0.887$) compared to both `CustomRandomForest` ($F1 = 0.939$) and `RandomForestClassifier` ($F1 = 0.936$). Furthermore, our improved model, `ImprovedGradientBoosting`, outperformed our preliminary implementation considerably on all metrics ($F1 = 0.939, \text{ROC AUC} = 0.936$)

Table 3: Classification Performance of Custom Random Forest Model and Gradient Boosting

	TP	FP	TN	FN	F1	ROC AUC
sklearn `RandomForestClassifier`	265	42	436	14	0.940	0.969
`CustomRandomForest` with ADASYN	251	56	431	19	0.920	0.971
sklearn `GradientBoostingClassifier`	**279**	**28**	**440**	**10**	**0.959**	**0.981**
`CustomGradientBoosting` with ADASYN	246	61	419	31	0.901	0.931
`ImprovedGradientBoosting` with ADASYN	273	34	439	11	**0.951**	0.965

After optimizing hyperparameters, the gradient boosted random forest models demonstrated superior performance compared to the basic random forest models across both F1 score and ROC AUC metrics. Between our custom methods, `ImprovedGradientBoosting` achieved the second highest F1 score of 0.951, which marginally underperformed the baseline sklearn `GradientBoostingClassifier` ($F1 = 0.959$).



(a) Confusion matrix

(b) ROC Curve

Figure 2: Best custom classification model: `ImprovedGradientBoosting`

Overall, the sklearn implementation of `GradientBoostingClassifier` had the best performance overall. However, our implementation of `ImprovedGraidentBoosting` on the ADASYN training and testing samples had very high classification performance overall. It appears that the sklearn model is slightly better at classifying the HL samples (our model has 6 more misclassified false positives).

4 DISCUSSION

In order to improve early-identification screenings for autism spectrum disorder (ASD) we implemented random forest models to predict likelihood of an ASD diagnosis. We leverage Population Phenoscreening, a large dataset that solves many of the challenges of existing machine learning implementations for classification in ASD samples. Namely, models are plagued by small sample sizes in diagnoses and inconsistent instruments administered between candidates which limits usable features for the models. Prior studies that implement machine learning models in this domain implement SVM, random forest, and naive bayes, but most of these studies focus on older samples (at which incidence rates of ASD are higher). This is a novel dataset which is not been extensively investigated, and has never had any machine learning models fit.

We implemented three variations of oversampling to address class imbalances: SMOTE, BorderlineSMOTE, and ADASYN. We hypothesized that BorderlineSMOTE would be the best performer, optimizing oversampling for separable samples within our HL class. However, model results suggest that ADASYN had the highest classification scores (F1 $= 0.939$) compared to both our implementation of `CustomRandomForest` (F1 $= 0.936$) and sklearn's `RandomForestClassifer` (F1 $= 0.936$). Researching the BorderlineSMOTE algorithm in more detail, we believe this result was due to high level of overlap between some HL and LL samples, and potentially due to outliers in the HL class. The ADASYN method specifically oversamples cases where there is overlap between the HL and LL classes, i.e., in regions of lots of heterogeneity, which is likely why this method improves metrics. We do note that the differences and improvement between these approaches is not considerable, and most differ only by a few cases. We see this also in the variation between ROC AUC scores, which actually is highest for the base `CustomRandomForest`.

After implementing PCA for this dataset, reduced dimensionality didn't improve overall classifications. There are two likely reasons why PCA didn't improve overall fits. First, PCA assumes linear relationships between variables. Second, while we normalized all of our metrics the distributions of each feature, this may not be the most clinically appropriate approach. As mentioned earlier in Section 2.4, some of the available features include the exact age when the instrument was administered. The better estimate of the input feature is realistically some z-scored value from a normative model that takes age into account. This would appropriately scale features in a clinical appropriate way, while normalizing scales for all metrics in a systematic and reasonable way that accounts for clinical variance in each measure (rather than assuming all features have normal distributions). Some subscores on instruments are scored and normed by age, but not all measures are. Alternatively, features could be selected directly from a clinician with context-specific intuition of subscore constructs to reduce the total number of features. Because model fits with PCA-scaled variables showed worse F1 scores than our ADASYN model and even `CustomRandomForest` model, and there were more valid methods to reduce features, we did not exhaustively pursue optimization for the scope of this project.

After implementing two custom models, `CustomRandomForest` and `ImprovedGradientBoosting`, we compared these models to the sklearn defaults `RandomForestClassifier` and `GradientBoostingClassifier` as a benchmark. From initial comparisons (see Table 1) it appeared that our `CustomRandomForest` model (F1 $= 0.939$), was outperforming the `RandomForestClassifier` (F1 $= 0.935$). However, after performing a full grid search to tune hyperparameters for each model (see Table 3), the `RandomForestClassifier` (F1 $= 0.940$) outperforms our implementation (F1 $= 0.920$). This jump in performance from the sklearn implementation may simply be due to the ADASYN oversampling, which in Table 1 comparisons we did not use explicitly in the sklearn model. Alternatively, it could also be due to overfitting in the sklearn model; we see that the best `n_estimator` value for `CustomRandomForest` is 30, compared to the 90 in the sklearn model reported in Table 2. This, in addition to a higher AUC ROC (0.971) suggests that our custom implementation may generalize to new datasets better.

The gradient boosted models, we implemented a rudimentary version of gradient boosting similar to what was discussed in class, in `CustomGradientBoosting`. Two primary improvements to this model made considerable gains in model performance: (1) we changed our loss from a simple residual to a log loss function and (2) we used `DecisionTreeClassifier` as the base classifier and added stochastic gradient descent to address heterogeneity within the dataset. Changing our loss function penalizes misclassification more severely when the model is confident about a misclassification. Changing to our decision tree classifier enables our models to more effectively capture nonlinear patterns in the dataset, and using stochastic gradient descent improves generalization of models since they are less likely to overfit the training data.

After performing a full grid search to tune hyperparameters for each model (see Table 3), the `RandomForestClassifier` (F1 $= 0.940$) outperforms our implementation (F1 $= 0.920$).

4.1 Future directions

This study identifies key areas to advance the use of machine learning in early-identification screenings for Autism Spectrum Disorder (ASD). Future work will address current limitations discussed in this project and enhance both model performance and clinical applicability.

For the data cleaning and preprocessing phase, the group could explore alternative approaches for handling missing data, such as using K-Nearest Neighbors (KNN) Imputation, which fills in missing values based on the similarity of nearby samples. Additionally, standardizing the dataset and employing feature selection techniques to retain only the most relevant features, using either an age-based z-score or simiply a clinically-driven data reduction method as previously discussed. We will also consider implementing outlier detection methods to identify and exclude anomalous data points. For dimensionality reduction, techniques like t-SNE, UMAP, Independent Component Analysis (ICA), and Linear Discriminant Analysis (LDA) could be applied to simplify the data while retaining its essential structure.

In the modeling stage, the group could experiment with advanced ensemble methods such as XGBoost, LightGBM, and CatBoost, which are well-suited for handling class imbalances and high-dimensional datasets. Incorporating neural network architectures, like feed-forward neural networks (FNNs) and convolutional neural networks (CNNs) could help capture complex, non-linear patterns in the data. Additionally, trying non-density-based and density-based clustering algorithms might reveal hidden structures or patterns within the dataset. To address unknown class labels, the group could explore semi-supervised learning techniques, which leverage both labeled and unlabeled data to improve classification accuracy.

For the context-specific analysis, we could delve deeper into feature exploration to assess how each feature influences model performance. This could provide valuable insights for refining and optimizing the model further, ensuring that the most impactful features are emphasized in the final implementation. Furthermore, this would be of novel clinical significance, because it would provide empirical evidence to determine which specific instruments should be administered for early-screening.

Beyond predicting ASD likelihood, future efforts will implement models for diagnosis prediction by incorporating the full longitudinal data. This approach will enable models to identify patterns leading to confirmed diagnoses, providing clinicians with tools for early intervention and outcome tracking.

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